

Introduction to Machine Learning

Course Notes — Module 1: Foundations

Machine learning is a field of artificial intelligence in which systems learn patterns from data rather than being explicitly programmed with rules. These notes cover the foundational concepts of supervised learning, unsupervised learning, overfitting, and model evaluation.

1. Supervised Learning

In supervised learning, the model is trained on a labeled dataset, where each input example is paired with a known correct output. The goal is for the model to learn a mapping function from inputs to outputs that generalizes well to new, unseen data.

Common supervised learning tasks include classification, where the output is a discrete category (for example, spam or not spam), and regression, where the output is a continuous numerical value (for example, predicting house prices).

Popular supervised learning algorithms include linear regression, logistic regression, decision trees, random forests, support vector machines, and neural networks.

2. Unsupervised Learning

Unsupervised learning works with unlabeled data. Instead of predicting a known output, the model tries to discover hidden structure or patterns within the data on its own.

The two most common unsupervised learning tasks are clustering, which groups similar data points together (for example, K-Means clustering), and dimensionality reduction, which reduces the number of features in a dataset while preserving as much important information as possible (for example, Principal Component Analysis, or PCA).

3. Overfitting and Underfitting

Overfitting occurs when a model learns the training data too closely, including its noise and random fluctuations, resulting in poor performance on new, unseen data. An overfit model typically shows very high accuracy on training data but significantly lower accuracy on test data.

Underfitting occurs when a model is too simple to capture the underlying pattern in the data, resulting in poor performance on both the training data and the test data.

Common techniques to reduce overfitting include cross-validation, regularization (such as L1 and L2 regularization), dropout in neural networks, pruning in decision trees, and collecting more training data.

4. Model Evaluation Metrics

For classification problems, common evaluation metrics include accuracy, precision, recall, and F1-score. Accuracy measures the overall proportion of correct predictions, while precision measures how many predicted positives were actually correct, and recall measures how many actual positives were correctly identified.

The F1-score is the harmonic mean of precision and recall, and is particularly useful when the class distribution is imbalanced.

For regression problems, common evaluation metrics include Mean Absolute Error (MAE), Mean Squared Error (MSE), and R-squared, which measures how well the model's predictions explain the variance in the target variable.

5. Train-Test Split and Cross-Validation

To evaluate how well a model generalizes to new data, the available dataset is typically split into a training set and a test set, commonly using an 80/20 or 70/30 split. The model is trained only on the training set and evaluated on the held-out test set.

K-fold cross-validation is a more robust evaluation technique in which the dataset is divided into K equal parts. The model is trained K times, each time using a different fold as the test set and the remaining folds as the training set. The final performance is the average across all K runs.